

FOUNDATIONS / METHODS

CONCEPTS RELATING TO RESEARCH DESIGN

2026—09—28

Dependent & independent variables



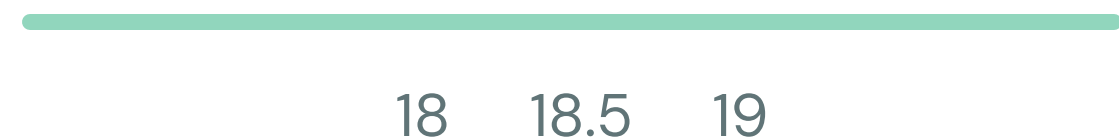
The dependent variable changes in response to one or more independent variables.

$$\text{height} = f(\text{age}, \text{sex})$$

Continuous vs non-continuous

CONTINUOUS

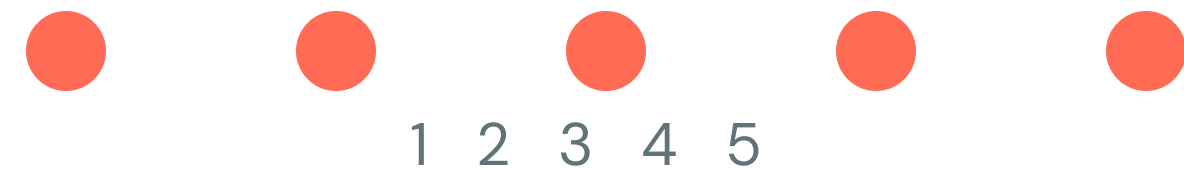
Any quantitative value, including decimals.
Example: age 18.5 years



- Phenomena which can take on quantitatively different values even in decimal points are called 'continuous variables'.

DISCRETE

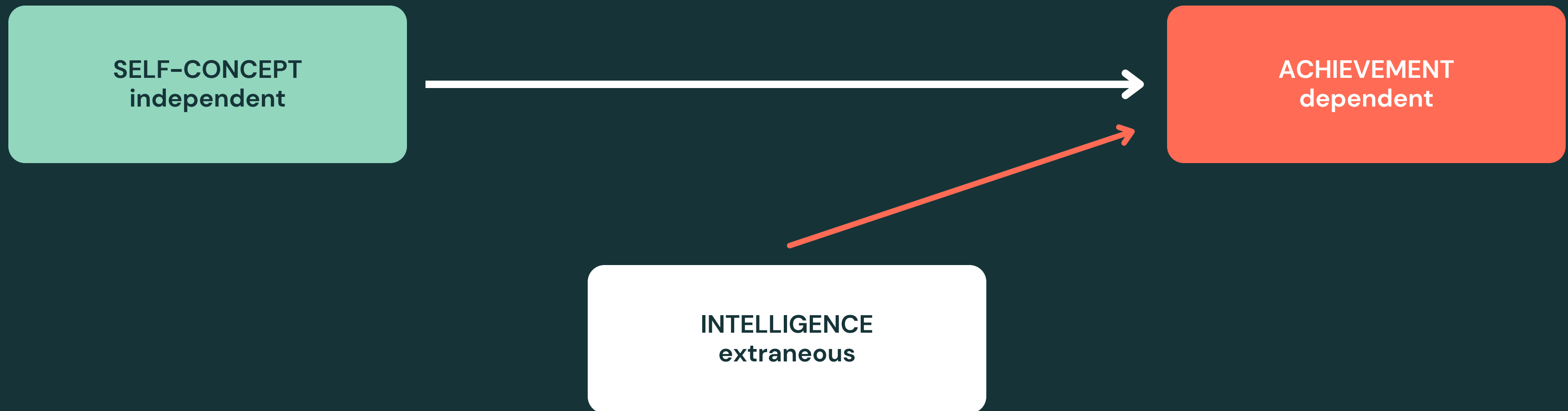
Whole-number counts only.
Example: 3 children



- If they can only be expressed in integer values, they are non-continuous variables or in statistical language 'discrete variables'.

Extraneous variable

An independent variable outside the study purpose that still affects the outcome.



- Independent variables that are not related to the purpose of the study, but may affect the dependent variable are termed as extraneous variables.

Control

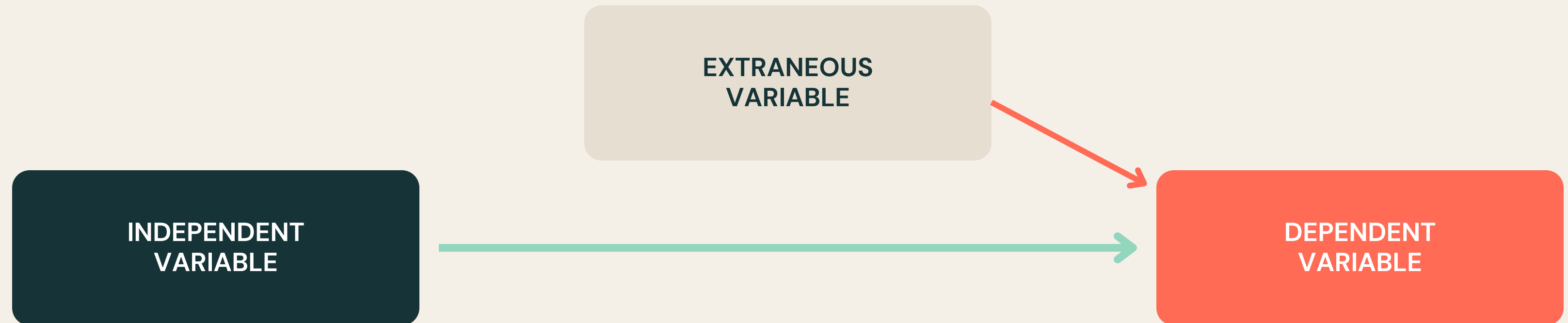
- One important characteristic of a good research design is to minimize the influence or effect of extraneous variable(s).

The technical term 'control' is used when we design the study minimizing the effects of extraneous independent variables.

In experimental researches, the term 'control' is used to refer to restrain experimental conditions.



Confounded relationship



- When the dependent variable is not free from the influence of extraneous variable(s), the relationship between the dependent and independent variables is said to be confounded by an extraneous variable(s).
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Research hypothesis

**IF X CHANGES,
THEN Y CHANGES.**

A testable predictive statement linking at least one independent variable to at least one dependent variable.

Assumptions and non-testable relationships do not qualify.

Two ways to test a hypothesis

EXPERIMENTAL

The researcher manipulates the independent variable, then observes the dependent variable.

NON-EXPERIMENTAL

The independent variable is observed as it occurs
— no manipulation is introduced.

Experimental & control groups

EXPERIMENTAL GROUP
novel / special condition

when the group is exposed to some novel or special condition, it is termed an 'experimental group'

CONTROL GROUP
usual condition

- In an experimental hypothesis-testing research when a group is exposed to usual conditions, it is termed a 'control group'

Treatments

01

FERTILIZER A

02

FERTILIZER B

03

FERTILIZER C

Treatments are the distinct conditions applied to groups — for example, three fertilizer varieties in a wheat–yield experiment.

Experiment

The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.

Experiments can be of two types viz., absolute experiment and comparative experiment.



Diagram illustrating the two types of experiments:

- A dark teal rounded rectangle on the left containing the word "ABSOLUTE" in white capital letters.
- A light teal rounded rectangle on the right containing the word "COMPARATIVE" in dark teal capital letters.
- The word "And" in red italicized font is positioned between the two rectangles.

ABSOLUTE

And

COMPARATIVE

Experimental unit(s)

Pre-determined plots or blocks where treatments are applied. Careful selection and precise definition protect the experiment's logic.

